Chapter 10: Language and Cognition Mathematical Equations

Comprehensive Neuroscience Reference

Abstract

This document presents the mathematical framework underlying language and cognition neuroscience. We develop equations for speech acoustics (spectral analysis, formant frequencies, phoneme confusion), dual-stream language models (ventral stream sound-to-meaning, dorsal stream auditory-motor), reading models (lexical and sublexical routes, frequency effects), executive function (working memory dopamine modulation, conflict monitoring), decision-making (drift diffusion accumulation, value-based choices, moral judgment pathways), creative cognition (network interaction), numerical magnitude representation (Weber's law, discrimination), and music timing (beat anticipation, tempo effects). Six detailed case studies with numerical simulations ground the theory in quantitative predictions from behavioral and neuroimaging data.

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1. Speech Acoustics and Phonological Processing

1.1 Spectral Analysis of Speech — Formant Frequencies

Speech is produced by modulating airflow through the vocal tract (mouth, lips, nasal cavity). The power spectrum reveals resonances (formants) that depend on vocal tract shape:

$$P(f) = |X(f)|^2 = \left| \int_{-\infty}^{\infty} s(t) e^{-i2\pi ft} dt \right|^2 \tag{1}$$

$$f_n = n \frac{c}{4L} \quad \text{(Simplified formant model)} \tag{2}$$

Symbol	Meaning	Value
$P(f)$	Power spectral density	dB/Hz
$s(t)$	Speech signal over time	Pressure waves (Pa)
f_n	Frequency of the n-th formant	Hz
c	Speed of sound in vocal tract	350 m/s
L	Effective vocal tract length	17–18 cm (males), 15–16 cm (females)
F1 (vowel height)	Varies inversely with tongue height	700–1220 Hz
F2 (vowel frontness)	Front-back tongue position	600–2500 Hz
F3 (rounding)	Rounded vs. unrounded vowels	2500–3500 Hz

The first and second formants (F1, F2) are most diagnostic for vowel identity. The superior temporal sulcus (STS) in the language-dominant hemisphere contains neurons selective for formant frequencies and transitions, forming the basis of phonological processing.

1.2 Phonological Distinctiveness and Phoneme Confusion

$$d_{\text{phon}} = \sqrt{(F_1^A - F_1^B)^2 + (F_2^A - F_2^B)^2} \tag{3}$$

$$P(\text{phoneme A perceived as B}) = \frac{1}{1 + e^{\beta d_{\text{phon}}}} \tag{4}$$

Phonological distinctiveness determines discrimination ease. Vowels far apart in formant space are easily distinguished; vowels close together cause confusion. This logistic function captures the psychometric relationship between acoustic distance and perceptual confusion.

2. Language Dual-Stream Model

2.1 Ventral Stream — Sound-to-Meaning Transformation

$$M = f_{\text{ventral}}(S) = \text{STS}(S) \xrightarrow{\text{decode}} \text{STG/MTG}(S) \xrightarrow{\text{semantic}} \text{ATL}(S) \tag{5}$$

$$\text{Activation}(S, t) = A_0 e^{-t/\tau_{\text{phoneme}}} \times G(S) + A_{\text{semantic}} \times G(\text{meaning}) \tag{6}$$

Symbol	**Meaning**	**Typical Value**
M	Semantic/meaning representation from sound	Multidimensional vector
STS	Superior temporal sulcus (phonological)	70–100 ms post-stimulus
STG/MTG	Superior/middle temporal gyrus (lexical-semantic)	150–300 ms
ATL	Anterior temporal lobe (combinatorial semantics)	300+ ms
τ_{phoneme}	Time constant for phoneme processing	100–200 ms

The ventral stream performs hierarchical sound-to-meaning mapping. Early neurons are sensitive to formant transitions and phonological features; further along, lexical-semantic information emerges; finally, the anterior temporal lobe combines word meanings into sentence-level representations.

2.2 Dorsal Stream — Sound-to-Motor Mapping

$$M_{\text{motor}} = f_{\text{dorsal}}(S) = \text{STS}(S) \xrightarrow{\text{phoneme}} \text{Spt}(S) \xrightarrow{\text{motor}} \text{IFG}(S) \tag{7}$$

$$\text{Motor activation}(t) = \int_0^t K(t - \tau) \cdot S(\tau) d\tau \quad \text{(Motor prediction via convolution)} \tag{8}$$

The dorsal stream maps speech sounds to motor commands for speech production (repetition, speaking). The Sylvian parietal-temporal area (Spt) acts as a sensorimotor interface, receiving phonological information and projecting to Broca's area to drive articulation. Damage to Spt or arcuate fasciculus causes conduction aphasia: patients can understand and speak but cannot repeat.

3. Reading and the Dual-Route Model

3.1 Lexical Route Activation

$$A_{\text{lexical}}(t) = A_{\max}(1 - e^{-t/\tau_{\text{lex}}}) \times S(\text{frequency}) \times S(\text{length}) \tag{9}$$

$$S(\text{frequency}) = \frac{\log(\text{Freq} + 1)}{\log(F_{\max})} \quad \text{(Frequency scaling)} \tag{10}$$

$$S(\text{length}) = \frac{1}{1 + 0.1 \times (\text{Length} - 5)} \quad \text{(Length penalty)} \tag{11}$$

Symbol	**Meaning**	**Value**
$A_{\text{lexical}}(t)$	Activation of lexical pathway	0–1
τ_{lex}	Time constant for lexical access	150–250 ms
Freq	Word frequency (occurrences per million)	1–10,000+
Length	Word length (letters)	3–15 typical

The lexical route is fast, automatic, and privileged for familiar, high-frequency words. The visual word form area (VWFA) in left fusiform gyrus rapidly activates whole-word representations. Frequency and length effects reflect learned orthographic-semantic associations.

3.2 Sublexical Route Activation

$$A_{\text{sublexical}}(t) = A_{\max}(1 - e^{-t/\tau_{\text{sub}}}) \times C(\text{regularity}) \tag{12}$$

$$C(\text{regularity}) = \begin{cases} 1.0 & \text{if phonologically regular} \\ 0.3 & \text{if phonologically irregular} \end{cases} \tag{13}$$

$$\text{Pronunciation latency} = \tau_{\text{sub}} + \Delta t_{\text{novelty}} + \Delta t_{\text{GPC}} \tag{14}$$

The sublexical (phonological) route converts orthography to sound via grapheme-phoneme conversion rules (left IPS/IPL, Spt region). This route is slower but flexible — it can pronounce novel words and pseudowords. Dyslexic individuals show reduced activation and connectivity in this route.

4. Executive Function and Working Memory

4.1 Dopamine Modulation of Working Memory

Prefrontal cortex (dlPFC) relies on dopamine D1 receptor stimulation for working memory maintenance:

$$\text{Persistent firing} = f_{\max} \times \frac{[D1]}{K_d + [D1]} \times (1 - \text{Distraction}) \tag{15}$$

$$\text{Performance} = f([D1]) = \text{baseline} + \text{slope} \times [D1] - \text{quadratic} \times [D1]^2 \tag{16}$$

This inverted-U relationship reflects:

- Too low dopamine: Weak D1 stimulation → poor sustained firing → working memory failures (ADHD, fatigue)
- Optimal dopamine: Maximum persistent neural activity → excellent working memory
- Too high dopamine: Excessive noise in firing patterns → working memory impairment (stress, psychosis)

4.2 Conflict Monitoring and Cognitive Control

The anterior cingulate cortex (ACC) detects conflict between competing responses:

$$\text{Conflict signal} = |A_{\text{response 1}} - A_{\text{response 2}}| \tag{17}$$

This conflict signal recruits dorsolateral PFC (dlPFC) to implement top-down control:

$$\text{Control strength} = \alpha \times \text{Conflict signal} + \text{Prior control level} \tag{18}$$

Example: Stroop task (name color of word "RED" printed in blue) — reading response (high automatic activation) competes with color-naming response (lower activation) → ACC detects conflict → dlPFC increases color-naming gain, suppresses reading.

5. Decision-Making

5.1 Drift Diffusion Model

Evidence for each option accumulates over time until a decision threshold is reached:

$$dx = v\,dt + \sigma\,dW_t \tag{19}$$

where v is drift rate (evidence strength), σ is noise magnitude, and dW_t is a Wiener increment.

First-passage time (decision latency) to threshold Θ:

$$T_{\text{mean}} = \frac{\Theta - x_0}{v} \quad \text{(For zero-noise approximation)} \tag{20}$$

Accuracy-speed tradeoff is controlled by threshold:

- High threshold: Slower but more accurate decisions
- Low threshold: Faster but more error-prone decisions

Neural accumulation occurs in lateral intraparietal area (LIP) for perceptual decisions and ventral striatum/vmPFC for value-based decisions.

5.2 Value-Based Decision-Making

Orbitofrontal cortex (OFC) encodes subjective value of options in a common currency:

$$\text{Value} = \text{OFC} = w_{\text{taste}} \times T + w_{\text{risk}} \times R + w_{\text{social}} \times S + w_{\text{expected}} \times E \tag{21}$$

where weights w_i are learned and can be modified by experience, context, and emotional state.

Choice probability follows logistic function:

$$P(\text{choose option A}) = \frac{1}{1 + e^{-\beta(\text{Value}_A - \text{Value}_B)}} \tag{22}$$

where β controls decision temperature (sensitivity to value differences).

5.3 Moral Decision-Making

Personal (emotional) moral dilemmas activate vmPFC/amygdala; impersonal (cognitive) dilemmas activate dlPFC/parietal cortex:

$$A_{\text{emotional}} = (\text{Harm severity}) \times \text{vmPFC responsivity} \times (1 - \text{Cognitive override}) \tag{23}$$

$$A_{\text{utilitarian}} = \frac{\text{Lives saved}}{\text{Lives harmed}} \times \text{dlPFC analysis strength} \tag{24}$$

$$\text{Final judgment} = (1 - w) \times A_{\text{emotional}} + w \times A_{\text{utilitarian}} \tag{25}$$

where w weights emotional vs. utilitarian components. Patients with vmPFC lesions have abnormally high w (utilitarian-biased).

6. Creativity and Neural Networks

6.1 DMN-ECN Coupling in Creativity

Highly creative individuals show atypical **positive coupling** between default mode network (DMN) and executive control network (ECN):

$$r_{\text{DMN-ECN}} = \rho \times \text{Creativity score} - \text{baseline anticorrelation} \tag{26}$$

Normal individuals: $r_{\text{DMN-ECN}} < 0$ (networks are anti-correlated) Creative individuals: $r_{\text{DMN-ECN}} > 0$ (networks are co-activated)

This enables simultaneous idea generation (DMN) and evaluation (ECN), explaining why creative people can brainstorm effectively.

7. Numerical Cognition

7.1 Weber's Law for Numerosity

The intraparietal sulcus (IPS) encodes approximate numerosity. Discrimination follows Weber's law:

$$\text{JND} = w \times n \quad \text{(Just-noticeable difference)} \tag{27}$$

where w is Weber fraction (∼0.15, or 15

Psychometric function:

$$P(\text{correct: } n_2 > n_1) = \Phi\left(\frac{\log(n_2/n_1)}{\sigma}\right) \tag{28}$$

where σ is the noise in the magnitude representation ($\sigma \approx 0.20$ in log units).

7.2 Triple-Code Model of Number

Numbers are represented in three complementary codes:

$$\text{Number} = \begin{cases} \text{Analog magnitude (IPS)} & \text{approximate, nonverbal} \\ \text{Visual Arabic numeral (fusiform)} & \text{visual recognition} \\ \text{Verbal code (perisylvian)} & \text{number words, facts} \end{cases} \tag{29}$$

Arithmetic facts (multiplication tables) are retrieved from left angular gyrus via verbal memory, while magnitude operations (comparison, estimation) engage bilateral IPS.

8. Music and Beat Tracking

8.1 Temporal Anticipation and Beat Prediction

The brain anticipates musical beats via cerebellar and basal ganglia mechanisms:

$$\sigma^2_{\text{beat}} = \sigma_0^2 + (w \times IOI)^2 \quad \text{(Beat timing variance)} \tag{30}$$

where IOI is inter-onset interval (time between beats), w is Weber fraction for timing ($\sim$0.15), and σ_0 is baseline variance.

Optimal beat anticipation:

$$t_{\text{anticipatory}} = IOI - (80\text{–}100 \text{ ms}) \tag{31}$$

Motor cortex activity (measured via motor evoked potentials, MEP) peaks 100 ms before the next beat.

Probability of synchronization within acceptable window (e.g., $\pm$100 ms):

$$P(\text{sync}) = \Phi\left(\frac{100}{\sigma_{\text{beat}}}\right) - \Phi\left(\frac{-100}{\sigma_{\text{beat}}}\right) \tag{32}$$

Slower tempos have wider timing windows (lower accuracy); faster tempos have tighter windows (higher accuracy but more demanding).

9. Case Studies and Simulations

9.1 Case Study 1: Stroop Task — Conflict Monitoring

Scenario: Model ACC conflict signal and dlPFC control strength during congruent vs. incongruent trials.

Given:

- Congruent trial (RED written in red): Reading activation 0.9, Color naming 0.85
- Incongruent trial (RED in blue): Reading 0.9, Color naming 0.3
- ACC sensitivity to conflict: $\alpha = 0.5$
- dlPFC control strength: baseline 0.1, increases with conflict

Congruent trial:

Conflict $= |0.9 - 0.85| = 0.05$ (minimal conflict)

Control strength $= 0.5 \times 0.05 + 0.1 = 0.125$ (minimal control needed)

Result: Fast, accurate color naming (200–250 ms)

Incongruent trial:

Conflict $= |0.9 - 0.3| = 0.6$ (strong conflict)

Control strength $= 0.5 \times 0.6 + 0.1 = 0.4$ (robust control engagement)

Result: Slower color naming (600–800 ms), high error rate without sufficient dlPFC control

Interpretation: ACC detects conflict and recruits dlPFC; this adaptive control reduces errors but increases latency on incongruent trials (classic Stroop effect).

9.2 Case Study 2: Drift Diffusion for Perceptual Decision

Scenario: Model decision time and accuracy for random dot motion perception (motion direction: left vs. right).

Given:

- Coherence 25% (weak motion signal): drift $v = 1.5$ units/s
- Coherence 50% (strong signal): $v = 3.0$ units/s
- Noise: $\sigma = 1.0$
- Decision threshold: $\Theta = 3.0$
- Initial position: $x_0 = 0$

Weak signal (25% coherence):

Mean decision time: $T = 3.0/1.5 = 2$ seconds

Accuracy (fraction crossing correct threshold): $\sim$70%

Strong signal (50% coherence):

Mean decision time: $T = 3.0/3.0 = 1$ second

Accuracy: $\sim$90%

Interpretation: Stronger motion signals produce faster and more accurate decisions. The drift-diffusion framework quantitatively predicts both reaction time and accuracy as functions of signal strength.

9.3 Case Study 3: Dopamine Inverted-U in Working Memory

Scenario: Model working memory performance across dopamine levels in dlPFC.

Formula: Performance $= 60 + 40 \times [D1]/(K_d + [D1]) - 60 \times ([D1])^2/([D1]_{\max})^2$

where performance is in percentage correct (0–100%), and $[D1]$ is normalized dopamine concentration (0–1).

| $[D1]$ level | Performance (%) | State | |—|—|—| | 0.0 (no dopamine) | 60 | Baseline; poor WM | | 0.3 | 72 | Low dopamine; ADHD-like | | 0.5 (optimal) | 88 | Peak working memory performance | | 0.7 | 78 | High dopamine; noise increases | | 1.0 (excess) | 65 | Psychosis-like; excessive noise |

The inverted-U relationship explains: - ADHD: Insufficient dopamine → poor working memory - Stress: Excessive catecholamines → working memory impairment - Optimal performance: Moderate dopamine levels

9.4 Case Study 4: Moral Judgment — Trolley Dilemma

Scenario: Compare moral judgments in personal vs. impersonal dilemmas for normal vs. vmPFC-lesioned patients.

Given:

- Personal harm salience: 0.3 (moderate emotional impact)
- Utilitarian benefit (one person dies vs. five saved): 0.7
- vmPFC lesion: $w_{\text{utilitarian}} = 1.0$ (emotion suppressed)

Normal person, personal dilemma (push someone off bridge):

$A_{\text{emotional}} = 0.3 \times 0.8 = 0.24$ (strong emotional inhibition)

$A_{\text{utilitarian}} = 0.7$ (modest utilitarian reasoning)

Judgment: $P(\text{"Yes"}) = 1/(1+e^{-5(0.7-0.24)}) = 1/(1+e^{-2.3}) = 0.91 = 9.1\%$ (mostly reject harming person)

vmPFC-lesioned patient, personal dilemma:

$A_{\text{emotional}} = 0.3 \times 0.1 = 0.03$ (emotion suppressed by lesion)

$A_{\text{utilitarian}} = 0.7$ (unchanged)

Judgment: $P(\text{"Yes"}) = 1/(1 + e^{-5(0.7-0.03)}) = 1/(1 + e^{-3.35}) = 0.97 = 97\%$ (abnormally utilitarian)

Interpretation: vmPFC lesion shifts moral judgment toward purely utilitarian reasoning by removing emotional constraints on decision-making.

9.5 Case Study 5: Numerical Discrimination — Weber Fraction

Scenario: Calculate discrimination thresholds for different numerosities.

Given: $w = 0.15$, $\sigma = 0.20$ (log units)

For numerosity $n = 50$:

JND $= 0.15 \times 50 = 7.5$ (smallest distinguishable number is 57.5, ratio 1.15)

Discrimination accuracy for comparing 50 vs. 57.5:

$P(\text{correct}) = \Phi(\log(57.5/50)/0.20) = \Phi(0.140/0.20) = \Phi(0.70) = 75.8\%$ (threshold-level performance)

This Weber fraction law holds from infants to adults, and even in some animal species.

9.6 Case Study 6: Beat Tracking — Music Tempo Effects

Scenario: Predict beat synchronization accuracy at three different tempos.

Given:

- Slow (100 bpm): IOI = 600 ms
- Medium (120 bpm): IOI = 500 ms
- Fast (180 bpm): IOI = 333 ms
- $\sigma_0 = 40$ ms, $w = 0.15$

Beat timing variance:

| Tempo | IOI | σ_{beat} | |—|—|—| | Slow | 600 | $\sqrt{40^2 + (90)^2} = 98.5$ ms | | Medium | 500 | $\sqrt{40^2 + (75)^2} = 85$ ms | | Fast | 333 | $\sqrt{40^2 + (50)^2} = 64$ ms |

Synchronization accuracy (within ±100 ms):

| Tempo | Accuracy | Difficulty | |—|—|—| | Slow | 69% | High timing variability; hard to sync | | Medium | 76% | Optimal; best synchronization | | Fast | 88% | Tight timing; very precise sync possible |

This explains why 120 bpm (medium tempo) is optimal for dancing and why music in this range is most "groovy."

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